

GCP Cloud Architecture Playbook

1. Purpose

This playbook provides formal, cloud-specific guidance for designing, governing, and operating architectures on GCP. It covers Event-Driven, API-First, Microservices, Lakehouse, Streaming, AI Platform, and Modern Data Stack patterns.

2. Scope

This playbook applies to enterprise architects, data architects, platform engineers, and product engineering teams building cloud-native solutions.

3. Guiding Principles

- API-First and Schema-First design
- Event-driven by default
- Zero Trust security model
- Infrastructure as Code (IaC) mandatory
- Observability by design
- Data as a Product
- FinOps and cost governance

4. Cloud-Specific Reference Technology Stack

Compute: Compute Engine, GKE, Cloud Run

Storage: Google Cloud Storage

Streaming: Pub/Sub

Data: BigQuery, Dataproc, Data Fusion

AI/ML: Vertex AI

Governance: IAM, Cloud KMS, Cloud Logging

5. Target Reference Architectures

This section defines the target-state architectures for API, Event, Microservices, Lakehouse, Streaming, AI Platform, and Modern Data Stack within the cloud platform.

6. Cloud Landing Zone Model

Defines standardized account/project structure, networking, identity, security controls, and baseline services required before application workloads are deployed.

7. RACI (Roles & Responsibilities)

Defines clear ownership across Architecture, Platform, Data, Security, and Product teams for each layer of the architecture.

8. FinOps Model (Cost Governance)

Includes budgeting, cost allocation, tagging standards, chargeback/showback models, and automated cost monitoring and alerts.

9. SRE / Reliability Model

Defines SLAs, SLOs, SLIs, incident response processes, blameless postmortems, and chaos engineering practices.

10. Data Governance Operating Model

Covers data classification, lineage tracking, quality SLAs, retention policies, and stewardship roles.

11. API & Event Governance Model

Covers OpenAPI standards, schema registry rules, versioning policies, and deprecation timelines.

12. MLOps Operating Model

Defines model lifecycle management, experiment tracking, feature governance, drift monitoring, and automated retraining.

13. DR/BCP Strategy

Defines backup policies, cross-region replication, RTO/RPO targets, and failover procedures.

14. Security Threat Model & Controls

Covers threat modeling, IAM least privilege, encryption standards, network segmentation, and continuous security monitoring.

15. Project Approval Checklist

- ADR approved
- Target architecture reviewed
- Security assessment completed
- Cost model approved
- Governance controls in place
- SLOs defined
- DR plan documented

Cloud-Specific Landing Zone Details

Cloud Landing Zone Details - GCP (Cloud Foundation) Core Components: - Resource Hierarchy: Organization → Folders → Projects - Network: Shared VPC with centralized firewall and DNS - Identity: Cloud Identity + IAM with least privilege - Governance: Organization Policies, Policy Intelligence - Security: Security Command Center, VPC Service Controls - Logging: Centralized Cloud Logging and BigQuery exports - Automation: Terraform for project vending and baseline setup - Operations: Cloud Monitoring, Backup and DR

Detailed RACI (Per Architecture Layer)

Detailed RACI Table (Per Architecture Layer) Layer | Responsible | Accountable | Consulted | Informed ----- Cloud Foundation | Platform Team | Cloud Architect | Security, Network | Product Networking | Network Team | Cloud Architect | Security | Platform Security & IAM | Security Team | CISO | Cloud, Data | Product API Gateway | Platform Team | Integration Architect | Product | Data Microservices | Product Team | Engineering Lead | Platform | Security Event Streaming | Data Platform Team | Data Architect | Engineering | Product Lakehouse | Data Platform Team | Chief Data Architect | BI, AI | Security Modern Data Stack | Analytics Team | Data Architect | Product | Platform AI/ML Platform | Data Science Team | AI Lead | Data Platform | Product Observability | SRE Team | Platform Lead | Security | Product